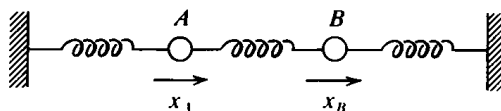


PHY201: Waves & Optics

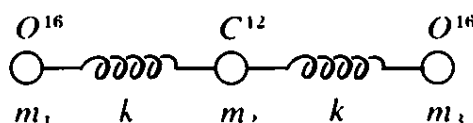
Problem Set 4

September 10, 2025

- Two identical pendulums are connected by a light coupling spring. Each pendulum has a length of 0.4 m, and they are at a place where $g = 9.8 \text{ m/sec}^2$. With the coupling spring connected, one pendulum is clamped, and the period of the other is found to be 1.25 sec exactly.
 - With neither pendulum clamped, what are the periods of the two normal modes?
 - What is the time interval between successive maximum possible amplitudes of one pendulum after one pendulum is drawn aside and released?
- Two harmonic oscillators A and B , of mass m and spring constants k_A and k_B , respectively, are coupled together by a spring of spring constant k_C . Find the normal frequencies ω' and ω'' and describe the normal modes of oscillation if $k_C^2 = k_A k_B$.
- Two equal masses on an effectively frictionless horizontal air track are held between rigid supports by three identical springs. The displacements from equilibrium along the line of the springs are described by coordinates x_A and x_B , as shown. If either of the masses is clamped, the period $T (= 2\pi/\omega)$ for one complete vibration of the other is 3 sec.



- If both masses are free, what are the periods of the two normal modes of the system? Sketch graphs of x_A and x_B versus t in each mode. At $t = 0$, mass A is at its normal resting position and mass B is pulled aside a distance of 5 cm. The masses are released from rest at this instant.
 - Write an equation for the subsequent displacement of each mass as a function of time.
 - What length of time (in seconds) characterizes the periodic transfer of the motion from B to A and back again? After one cycle, is the situation at $t = 0$ exactly reproduced? Explain.
- The CO_2 molecule can be likened to a system made up of a central mass m_2 connected by equal springs of spring constant k to two masses m_1 and m_3 (with $m_3 = m_1$).



- Set up and solve the equations for the two normal modes in which the masses oscillate along the line joining their centers. [The equation of motion for m_3 is $m_3(d^2x_3/dt^2) = -k(x_3 - x_2)$ and similar equations can be written for m_1 and m_2 .]

- (b) Putting $m_1 = m_3 = 16$ units, $m_2 = 12$ units, what would be the ratio of the frequencies of the two modes, assuming this classical description were applicable?
5. Two equal masses m are connected to three identical springs (spring constant k) on a frictionless horizontal surface (see figure). One end of the system is fixed; the other is driven back and forth with a displacement $X = X_0 \cos \omega t$. Find and sketch graphs of the resulting displacements of the two masses.

